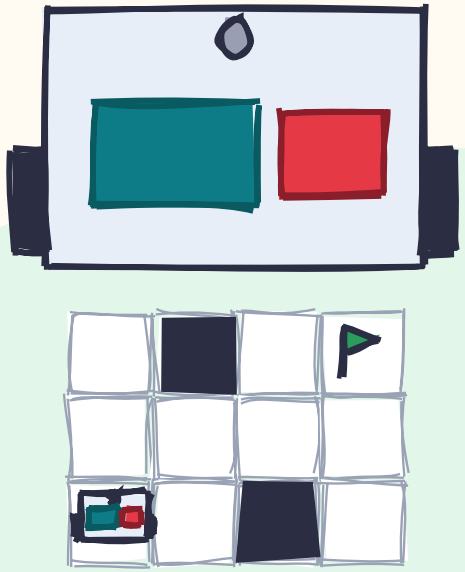
Little Explorers · Class 19

Program the Robot Car Through a Maze

Plan a path, snap the blocks, watch it drive!

Directions **Plan a path** **Code the car**



Did you know?
Robot vacuum cleaners plan paths round your furniture, all by themselves!

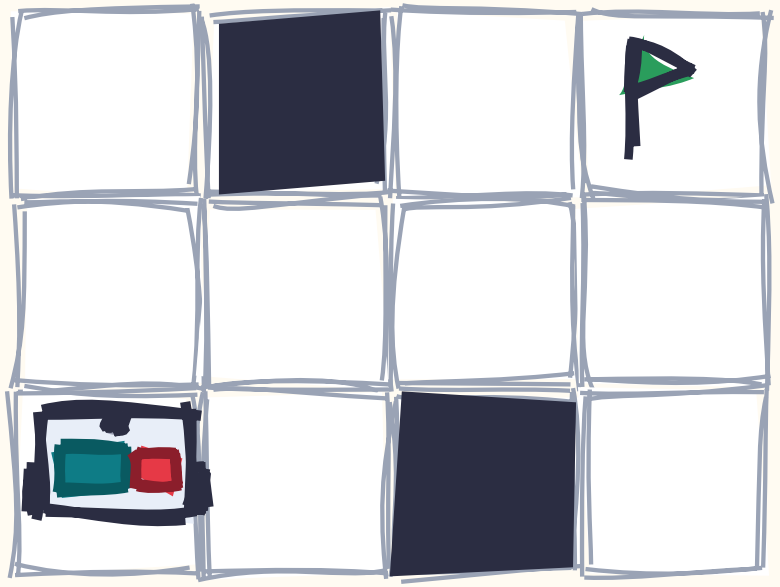
LEARN

Let's wonder...

How can a robot find its way?

A lost robot car

Our robot car wants to get to the flag, but there are walls in the way! It can't see the maze. It only knows what **we** tell it. What should we say?



Go!

Find the flag!

Step-by-step directions!

Today: you plan a path through a **floor maze** and program a real **robot car** to drive it.

Learning objectives

Today I can...

1	**Directions:** say **forward, back, left, right** and test them	**Proof:** I showed left and right with my arms
2	**Plan a path:** plan a path through a maze with **arrow cards**	**Proof:** my arrow cards lead to the goal
3	**Code the car:** snap picture blocks so the **robot car** follows my path	**Proof:** our car reached the goal

Objective 1 · Directions · video

Video time!



Fun Facts About Robots!

Nat Geo Kids · about 2 minutes · youtu.be/UuPAImipntw

Watch for...

1. Name one robot from the video.
2. How do robots **move**?
3. What jobs do robots help with?

Scan to watch

After the video: If you had a robot car, where would you send it in our classroom?

LEARN

Objective 1 · Directions

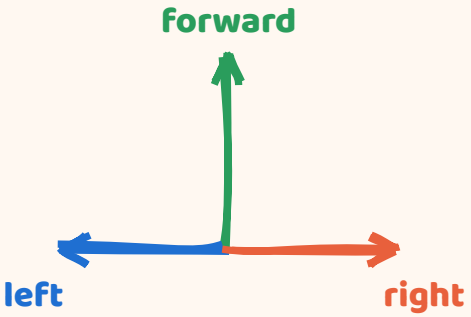
Four directions

The idea

Forward: go straight ahead. **Back**: go backwards.
Turn left and **turn right**: spin on the spot, then go forward.

Left and right are tricky! Left is the hand that makes an **L** with your thumb.

Example: a bus driver



A bus driver follows a route: straight, turn left at the shop, straight, turn right at the mosque. Robots need routes too!

Objective 2 · Plan a path

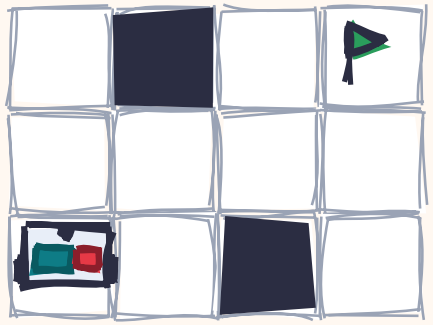
Plan with arrow cards

The idea

Before coding, plan on the floor! Lay arrow cards on each square from the car to the flag.

Count the squares: **forward 2, turn right, forward 1**. Walk the path with your finger to check.

Example: a treasure map



Pirate maps say 'walk 10 steps, turn at the big tree, dig!'. Our arrow cards are a treasure map for the robot.

Our machine

Feel, think, show



How it works

We plan the path.
The Arduino in the car **remembers** our blocks.
It turns the **motors** to drive: both wheels = forward, one wheel = turn.

Teacher: parts are pre-wired on the breadboard. Children plug in the USB cable, snap the picture blocks in mBlock and test.

Real world

Robots that find their way



Robot vacuums
Drive round furniture



Warehouse robots
Carry boxes along paths



Mars rovers
Drive on another planet!



Maps on phones
Plan a route for cars

? Where would you like a robot to drive to in our school?

Mars!
Rovers on Mars get their driving instructions from people on Earth, just like our car.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Simon says robot
2. Plan our path
3. Snap the picture blocks
4. Test drive!
5. Maze challenges

Class game

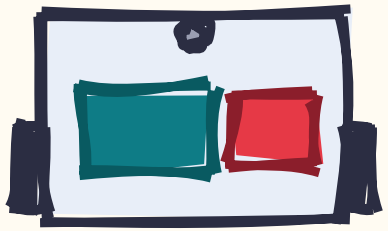
Simon says robot

How to play

1. The teacher is the programmer. Children are robots.
2. 'Robot says: forward 2 steps!' 'Robot says: turn left!'
3. Without 'robot says', don't move!
4. Last robot standing wins.

Then discuss

Which was harder to remember: left or right?



Big idea

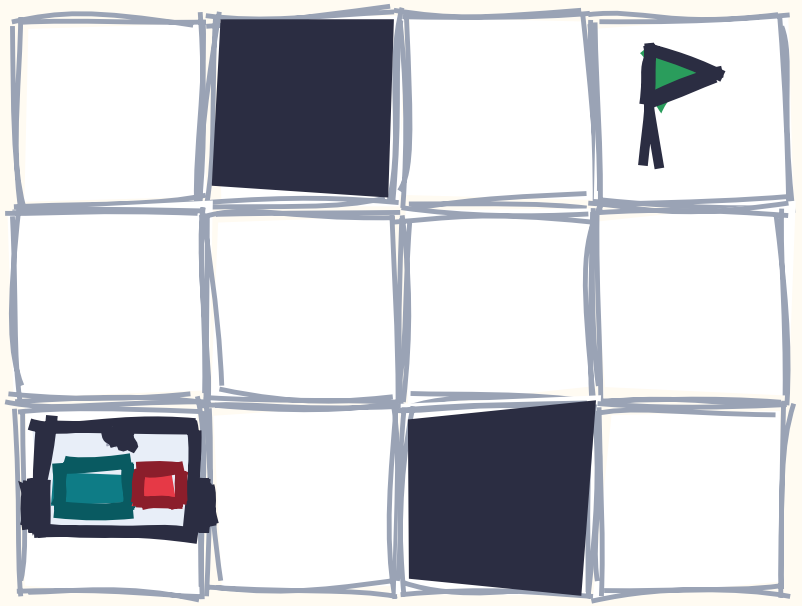
Robots only move when they get an instruction.

MAKE

Objective 2 · Plan a path · plan it

Plan our path

Step	Arrow	How many squares?
1		
2		
3		
4		
5		

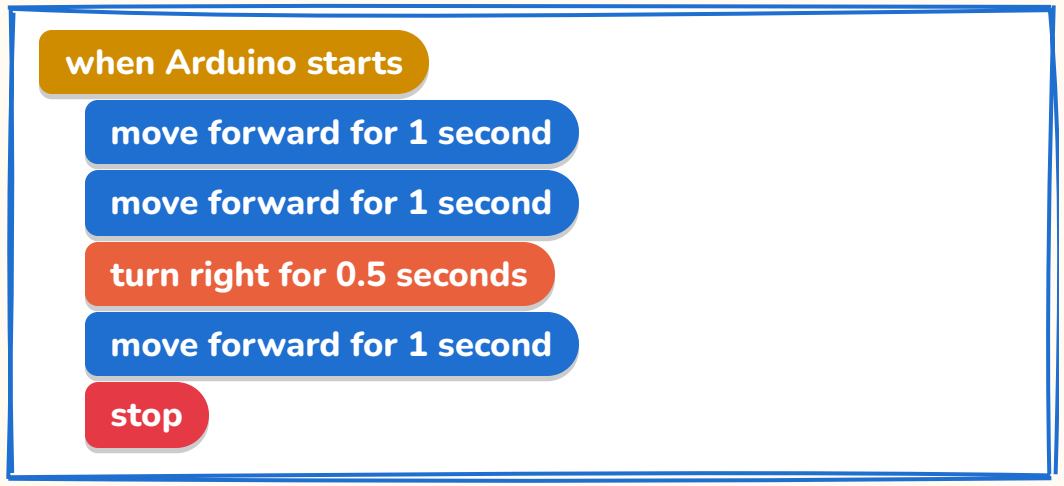


Check!
Walk the path with your finger before coding.

MAKE

Time to build!

Snap the picture blocks

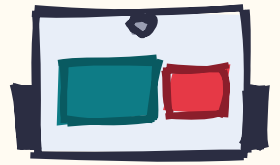


Read it like a story

Each block is one arrow card.

1 second of forward = about one square.

Turn, then forward!



Teacher

Upload with mBlock 5 using picture blocks. Test how long one square takes on your floor first!

Drive it!

Test drive!

Try	Reached the flag?	What we changed
Try 1	yes / no	_____
Try 2	yes / no	_____
Try 3	yes / no	_____

Talk about it

Did the car turn too much or too little? What number did you change?

Small changes

Change one number at a time: 0.5 → 0.6 seconds.

Level up!

Maze challenges

Choose your challenge



Harder maze: add more walls and plan again.



Loop it: use 'repeat 2 times: forward' instead of two blocks.



Beep: make the car beep when it reaches the flag.



Swap: drive another team's maze.

Think like an engineer

Why might the car go wrong on a slippery floor?

Boss level

Make the car drive there AND come back!

Objective check

Can I really do it? Prove it.

01	Show me left and right with your arms!	○ ○ ○
02	Show your arrow-card path.	○ ○ ○
03	Did your car reach the flag? What did you change?	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Robot car race: which team reaches the flag in the fewest tries?

Show what you know

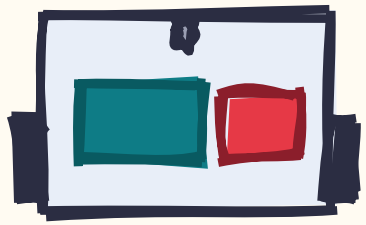
What did we learn?

- [x] Forward, back, left and right
- [x] Plan the path before coding
- [x] Each block is one step
- [x] Test, change one number, test again

Exit ticket: which way is left?

At home

Give someone at home directions from the door to your bed!



Next class: ScratchJr Dance Party!